

BSMS Computer Science candidate at RIT (May 2027) with 2× Paychex SWE co-ops. Built 5+ production MCP tools in Go and a 6-agent agentic AI pipeline deployed across 8 OpenShift microservices.

WORK EXPERIENCE

Software Engineering Co-op

Paychex, Inc., Rochester, NY | Jan 2026 – Aug 2026

- **Designed a 6-agent agentic AI pipeline** (GitHub Copilot CLI + MCP Protocol) that automated the full OpenShift Liveness/Readiness probe migration and OpenTelemetry adoption lifecycle end-to-end cutting per-service implementation time from **1–2 days to under 30 minutes**, deployed across 8 production microservices.
- **Designed an internal PR Review Bot** (Claude code via internal MaaS endpoint) integrated with Bitbucket/Jira/Confluence through MCP tooling that classifies changes and posts severity-ranked code review findings across a **10+** repo team.
- **Extended an internal developer AI platform** by building **5+ production-grade MCP tools in Go**: Webex (28 tools, OAuth 2.0), Oracle via custom TNS resolver, SQL Server, and Grafana Tempo; adopted by **50+ engineers** across the org.
- **Engineered a security-first data access layer**: PHI/PII redaction (36+ HIPAA patterns), SQL injection prevention, SELECT-only enforcement, and append-only audit logging validated with **128+ unit tests** across fully mocked infrastructure.
- Shipped **25+** stories across the full engineering stack: Oracle PL/SQL production releases, AQG Gates-as-Code YAML migration for 3 services, Automated Test migration from legacy to docker container based Robot Framework tests, and 20+ internal knowledge base pages.
- Scrum Master for my intern team (Python Rocks) where I lead weekly stand-ups, track sprint progress and manage Jira boards.

Security Analyst

Shaksy Engineering Services, Muscat, Oman | June 2025 – August 2025

- **Automated** infrastructure monitoring with **Python/SQL** and used **Wireshark/Nmap** on **Linux** to diagnose network issues, improving system visibility and documenting connectivity bottlenecks for senior engineers.

Software Engineering Co-op

Paychex, Inc., Rochester, NY | Jan 2024 – May 2024

- **Diagnosed and optimized Pro C and SQL query performance** across **4+ backend services**, reducing runtime errors by **35%** for a high-volume payroll platform serving millions of employees.
- **Authored 20+ automated unit and integration tests** integrated into CI/CD pipelines, improving release confidence and reducing regression risk across sprint deliverables.

ARTIFICIAL INTELLIGENCE

- Applied **structured agentic design** to eliminate LLM failure modes: typed **'AgentState' (TypedDict)** over freeform context, **deterministic conditional routing edges** over prompt based routing and structural **'interrupt_before()' gates** over instructional pauses - converting single-context-window-prompt loops into **crash-resilient directed StateGraphs**.
- Navigating the **enterprise AI agent deployment lifecycle**: **LangGraph** scaffolding, **LangSmith** trace observability, **LLM API key governance**, and **environment-gated CI/CD**.

SKILLS

- **Languages**: Java · Python · Go · C# · SQL · TypeScript · JavaScript · C++ · Bash
- **Frameworks & Tools**: LangGraph · LangSmith · Spring Boot · .NET · REST APIs · MCP · OAuth 2.0 · Docker · Git · Gradle · Jsonnet ·
- **Databases**: PostgreSQL · MongoDB · Oracle SQL · SQL Server · SQLite
- **Cloud & Infra**: OpenShift · Grafana Tempo · OpenTelemetry · Jenkins · AWS (exp) · Linux

EDUCATION

Rochester Institute of Technology (RIT) — *BS/MS Computer Science*

May 2027

- **Dean's List**: Fall '21, '23, '24, '25 · Spring '25
- **Focus Area**: Artificial Intelligence / Machine Learning and Data Science

PROJECTS

OpenMusic - Distributed Data Pipeline

PostgreSQL 15 · MongoDB 8.0 · Apache Spark / PySpark · Docker Compose · Python

- **Deployed a containerized relational-to-document migration pipeline**, ingesting **106M+ records** (PostgreSQL) and migrating 25.2M validated tracks into a read-optimized MongoDB schema - collapsing complex join queries from **9.11 minutes to O(1)** index lookups.
- **Debugged Spark OOM failures** during frequent itemset mining across 1.6M artist records by re-engineering from global distributed self-joins to **PySpark map transformations**, cutting network shuffle overhead and enabling linear scalability.

User-Space Keylogger Detection via OS Telemetry

Python · Scikit-Learn · Psutil · PyWin32 · Pynput · Random Forest

- **Built a real-time OS telemetry pipeline** sampling **17 normalized process metrics** (CPU, RSS/VMS memory, handles, threads, context switches), autonomously terminating malicious PIDs at a **≥ 90% confidence threshold** using a 200-tree class-balanced Random Forest.
- **Achieved 86% classification accuracy** on a 15:1 imbalanced dataset; identified `memory_rss`, `read_bytes`, `read_count`, and `num_handles` as top signals via tree impurity (Gini) analysis.